

Fine-Tuning Pipeline

Llama-3.1-8B-Instruct
Pre-trained base model (FP16)

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QLoRA Fine-Tuning
83K cyber examples | rank-16 | 2 epochs

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Fine-tuned Checkpoint (FP16)
WMDP ≈44-46% | MMLU ≈63-65%

9 quantized configurations (3 formats × 3 calibration conditions)

	FP8 E4M3	AWQ INT4	SQ INT8
C3 DACS (domain-aligned)	62.18%	59.64%	57.93%
C1 Generic (The Pile)	62.18%	59.42%	59.23%
C2 Self-gen (T=0.8)	62.18%	59.72%	49.92% (!)

Calibration × Format Matrix

Eval

WMDP
Cyber
1,987
MCQs

MMLU
14,042
MCQs